

Sukrit Singh

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Positions and Education

Damon Runyon Quant. Bio. Postdoctoral Research Fellow Memorial Sloan-Kettering Cancer Center – Damon Runyon Quantitative Biology Fellow – NCI K99/R00 Pathway to Independence Award Fellow – Member of the Folding@home consortium – Advisors: Dr. John Chodera & Dr. Markus Seeliger (Stony Brook University School of Medicine)	<i>2021 – present</i>
Ph.D., Computational and Molecular Biophysics Washington University in St. Louis – Thesis: <i>Understanding and exploiting protein allostery & dynamics using molecular simulations</i> – Advisor: Dr. Gregory R. Bowman	<i>2020</i>
B.A., Chemistry and Biology Washington University in St. Louis – Undergraduate thesis: <i>Synthesizing an amide bond nitroxide for improving intermolecular distance measurements.</i> – Research advisors: Dr. Garland R. Marshall & Dr. Jay Ponder	<i>2014</i>

Funding, Awards, & Fellowships

Global Research Fellow in AI & Precision Medicine (declined) Lund University, Sweden	<i>March 2026</i>
NCI Pathway to Independence Award for Outstanding Early-Stage Postdoctoral Researchers (K99/R00) National Cancer Institute, USA	<i>June 2024 – present</i>
Best Talk – NCI Junior Investigators meeting National Cancer Institute, Junior Investigators meeting	<i>August 2023</i>
Damon Runyon Quantitative Biology Fellowship Damon Runyon Cancer Research Foundation	<i>May 2022 – June 2024</i>
Community Ambassador eLife journal	<i>Jan. 2022 – Jan. 2023</i>
Millipore-Sigma Fellowship WUSM Dept. of Biochemistry and Molecular Biophysics	<i>March 2019 – 2020</i>
Best Poster – Runner up Biochemistry and Molecular Biophysics Department Retreat	<i>September 2016</i>
MCC Travel Award at UIUC Materials Computation Center at Univ. Illinois Urbana-Champaign	<i>September 2015</i>
WU Career Center Summer Internship Award Washington University in St. Louis	<i>June – August 2013</i>
HHMI Summer Undergraduate Research Fellowship Washington University in St. Louis	<i>June – August 2011</i>

Leadership & Service Positions

Funding Proposal Review Committee, <i>MSKCC AI and Innovations Technology Development Fund (aiTDF)</i>	Sept. 2025
Co-Chair, Gordon Research Seminar, <i>2027 Gordon Research Seminar (GRS) in Computer Aided Drug Design (CADD)</i>	2025 – present
Technical Advisory Committee, <i>OpenFreeEnergy Consortium</i>	Nov. 2024 – present
Organizer & Host, <i>Folding@NYC & NY Area Group for Informatics and Modeling (NYAGIM)</i>	July 2023 – present
Maintainer and Admin, <i>MDTraj project (github)</i> – MDTraj is an open-source library with >1 million downloads for analysis of molecular dynamics simulations – Manage community contributions, upkeep, and releases	April 2022 – present
Faculty Representative, Chodera lab <i>Folding@home consortium (website)</i> – Represent interests for John Chodera's group in the Folding@home consortium and coordinate billing	April 2022 – present
Managing director, Communications, Outreach, and Digital <i>Folding@home consortium (website)</i> – Manage social media (@foldingathome on twitter, ~32.0K followers) and outreach efforts (most heavily in 2020) – Run live streams on the Folding@home twitch channel	
Co-chair of Platform Session, <i>2025 Biophysical Society Meeting</i>	Feb. 2025
Trainee Advisory Committee member, <i>Living Journal of Computational Molecular Sciences</i>	Mar. 2021 – Mar. 2023
Chair of Student Liaison Committee and Friday Seminar Organizer <i>Department of Biochemistry and Molecular Biophysics</i> – Organized and ran the weekly Friday Work In Progress seminars – Ran yearly meetings of the committee with the department chair to discuss student issues, events, and activities	2017 – 2020

Scientific Publications

As of 04/2026, I have published **32** papers with over 2700 citations. My **h-index** is 18 and my **i10 index** is 21. My full publications list can be found on [Google Scholar](#), [MyNCBI bibliography](#), or [my website](#).

Author roles: + corresponding/senior author | * co-first | † lead computational

IN PREPARATION (Draft available upon request):

1. Kaminow, B., Payne, A.M., . . . , **Singh, S.⁺**, Chodera, J.D.⁺, *Physical biases improve performance of structure-based binding affinity models*, Pending submission.

AVAILABLE PUBLICATIONS:

1. Ranepura, G.A., Chowdhury, S.I., . . . , Chodera, J.D., **Singh, S.⁺**, Gunner, M.R.⁺, *Kinase inhibitors can change protonation or tautomeric state upon binding*, In Review, Available online on biorXiv; 2026.07.27.741060.
2. Zhu, Y., Degiacomi, M.T., Mey, A.S.J.S.⁺, **Singh, S.⁺**, *Tyrosine kinases sample unique activation ensembles*, In Review. Available online on biorXiv; 2026.06.26.734786.
3. Payne, A.M., Kaminow, B., . . . , Chodera, J.D.⁺, **Singh, S.⁺**, *How many crystal structures do you need to trust your docking results?*, In Press, *J. Med. Chem.*, 2026; Available on BioRxiv 2025.09.19.677428.
4. **Singh, S.⁺**, Clerkin, A., Castellanos, M.A., Chodera, J.D.⁺, *More protein-ligand data are needed for AlphaFold-like models to enable drug discovery*, *Current Opinions in Structural Biology*, 2025, Vol. 98, 103257.
5. **Singh, S.⁺**, Hanson, S.M.⁺, *Running and analyzing massively parallel molecular simulations*, In Press, *Methods in Mol. Bio.: Biomolecular Simulations*, 2025, Springer Nature.
6. **Singh, S.⁺**, Gapsys, V.^{*}, Aldeghi, M.^{*}, Schaller, D., White, J.B., Bluck, J., . . . , Volkamer, A., Christ, C., Seeliger, M.A., Chodera, J.D., *Prospective evaluation of physics-based simulation methods reveals their ability to classify the impact of kinase mutations on inhibitor binding*, *J. Phys. Chem. B*, 2025, 129(11), 2882–2902.
7. Chen, W., Gupta, A., . . . , **Singh, S.[†]**, Chodera, J.D., . . . , Chandarlapaty, S., *Trastuzumab Deruxtecan (T-DXd) Resistance via Loss of HER2 Expression and Binding*, *Cancer Discovery*, 2025.
8. Gerstberger, S., Hartner, S., . . . , **Singh, S.[†]**, . . . , Ganesh, K., *Pks+ E. coli triggers intestinal stem cell plasticity and early onset colorectal cancer*, Submitted, 2025.
9. López-Ríos de Castro, R., Rogríquez-Gerra, J., . . . , **Singh, S.**, Kramer, P.L., Pérez-Hernandez, G., Volkamer, A., Chodera, J.D., *Lessons learned during the journey of data: from experiment to model for predicting kinase affinity, selectivity, polypharmacology, and resistance*, *Living Journal of Comp. Mol. Sci.*, 2025, 6(1), 3875.
10. Eastman, P., . . . , **Singh, S.**, Swails, J., Turner, P., Wang, Y., Zhang, I., Chodera, J.D., De Fabritiis, G., Markland, T.E., *OpenMM 8: Molecular dynamics simulation with machine learning potentials*, *J. Phys. Chem. B*, 2024, 128(1), 109–116.
11. Nigam, A., Hurley, M.F.D., . . . , **Singh, S.**, . . . , Boura, E., Vedadi, M., Aspuru-Guzik, A., *Application of established computational techniques to identify potential SARS-CoV-2 Nsp14-MTase inhibitors in low data regimes*, *Digital Discovery*, 2024, 3, 1327–1341.
12. Zhang, I., Rufa, D.A., Pulido, I., . . . , **Singh, S.⁺**, Chodera, J.D.⁺, *Identifying and overcoming the sampling challenges in relative binding free energy calculations of a model protein:protein complex*, *J. Chem. Theory Comput.*, 2023, 19(15), 4863–4882.
13. Perner, F., Stein, E.M., Wenge, D.V., **Singh, S.[†]**, . . . , Chodera, J.D., Nowak, R.P., Fischer, E.S., Levine, R.L., Armstrong, S.A., Cai, S.F., *MEN1 mutations mediate clinical resistance to Menin inhibition*, *Nature*, 2023, 615, 913–919.
14. Outhwaite, I.R., **Singh, S.**, Berger, B.-T., Knapp, S., Chodera, J.D., Seeliger, M.A.,

Death by a thousand cuts – Combining kinase inhibitors for selective target inhibition and rational polypharmacology, *eLife*, 2023, 12:e86189.

15. Bobby, M., ..., **Singh, S.**, ..., *Open Science Discovery of Oral Non-Covalent SARS-CoV-2 Main Protease Inhibitor Therapeutics*, *Science*, 2023, 382(6671).

PUBLICATIONS FROM PHD WORK AND PRIOR (WASH. U.):

16. Vithani, N., Todd, T.D., **Singh, S.**, Blumer, K.J., Bowman, G.R., *G protein activation occurs via a largely universal mechanism*, *J. Phys. Chem. B*, 2024, 128(15), 3554–3562.
17. Todd, T.D., Vithani, V., **Singh, S.**, Bowman, G.R., Blumer, K.J., Soranno, A., *Stabilization of interdomain closure by a G protein inhibition*, *Proc. Nat. Acad. Sci.*, 2024, 121(36), e2311711121.
18. Cruz, M.A., Frederick, T.E., Mallimadugula, U.L., **Singh, S.[†]**, ..., Bowman, G.R., *A cryptic pocket in Ebola VP35 allosterically controls RNA binding*, *Nature Communications*, 2022, 13, 2269.
19. Knoverek, C.R., Mallimadugula, U.L., **Singh, S.[†]**, ..., Kay, L.E., Bowman, G.R., *Opening of a cryptic pocket in β -lactamase increases penicillinase activity*, *Proc. Nat. Acad. Sci.*, Nov 2021, 118(47), e2106473118.
20. Vithani, N., Ward, M.D., ..., **Singh, S.**, Bowman, G.R., *SARS-CoV-2 Nsp16 activation mechanism and a cryptic pocket with pan-coronavirus antiviral potential*, *Biophysical Journal*, 2021, 120(14), 2880–2889.
21. Zimmerman, M.I., Porter, J.R., Ward, M.D., **Singh, S.**, ..., Chodera, J.D., Bowman, G.R., *SARS-CoV-2 Simulations Go Exascale to Capture Spike Opening and Reveal Cryptic Pockets Across the Proteome*, *Nature Chemistry*, 2021, 13, 651–659.
22. Cubuk, J., Alston, J.J., Incicco, J.J., **Singh, S.[†]**, ..., Bowman, G.R., Hall, K.B., Soranno, A., Holehouse, A.S., *The SARS-CoV-2 nucleocapsid protein is dynamic, disordered, and phase separates with RNA*, *Nature Communications*, 2021, 12, 1936.
23. Brown, C.A., Hu, L., Sun, Z., Patel, M.P., **Singh, S.[†]**, ..., Bowman, G.R., Palzkill, T.M., *Antagonism between substitutions in β -Lactamase explains a path not taken in the evolution of bacterial drug resistance*, *Journal of Molecular Biology*, 2020, 295(21), 7376–7390.
24. **Singh, S.***, Sun, X.* , Blumer, K.J., Bowman, G.R., *Simulation of spontaneous G protein activation reveals a new intermediate driving GDP unbinding*, *eLife*, 2018, 7:e38465.
25. Reddy, D.N., **Singh, S.[†]**, Ho, C.M.W., ..., Doctor, A., Marshall, G.R., *Design, synthesis, and biological evaluation of stable $\beta^{6,3}$ -Helices: Discovery of non-hemolytic antibacterial peptides*, *Eur. J. Med. Chem.*, 2018, 149, 193–210.
26. Patrick, G.J., Fang, L., Schaefer, J., **Singh, S.[†]**, Bowman, G.R., Wencewicz, T.A., *Mechanistic Basis for ATP-Dependent Inhibition of Glutamine Synthetase by Tabtoxinine- β -Lactam*, *Biochemistry*, 2017, 57(1), 117–135.
27. **Singh, S.**, Bowman, G.R., *Quantifying allosteric communication via both concerted structural changes and conformational disorder with CARDS*, *J. Chem. Theory Comput.*, 2017, 13(4), 1509–1517.
28. Cascella, B., Lee, S.G., **Singh, S.[†]**, Jez, J.M., Mirica, L.M., *The small molecule JIB-04 disrupts O₂ binding in the Fe-dependent histone demethylase KDM4A/JMJD2A*, *Chem. Commun.*, 2017, 53, 2174–2177.

29. Brosey, C.A., Ho, C.M.W., Long, W.Z., **Singh, S.**, . . . , Bowman, G.R., Ellenberger, T.E., Tainer, J.A., *Defining NADH-Driven Allostery Regulating Apoptosis-Inducing Factor, Structure*, 2016, 24, 2067–2079.
 30. O'Connor, S.D., Cabrera, O.H., Dougherty, J.D., **Singh, S.**, . . . , Farber, N.B., Noguchi, K.K., *Dexmedetomidine protects against glucocorticoid induced progenitor cell apoptosis in neonatal mouse cerebellum*, *J. Matern. Fetal. Neonatal. Med.*, 2017, 30, 2156–2162.
 31. Nelson, C.A., Epperson, M.L., **Singh, S.**[†], Elliott, J.I., Fremont, D.H., *Structural Conservation and Functional Diversification within the Poxvirus Immune Evasion (PIE) Domain Superfamily*, *Viruses*, 2015, 7, 4878–4898.
 32. Cabrera, O.H., Dougherty, J.D., **Singh, S.**, Swiney, B.S., Farber, N.B., Noguchi, K.K., *Lithium protects against glucocorticoid induced neural progenitor cell apoptosis in the developing cerebellum*, *Brain Research*, 2014, 1545, 54–63.
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Invited Talks

As of 06/2026, I have given **32 invited seminars** worldwide.

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| 1. Telluride Workshop in Protein Electrostatics
<i>Telluride, CO, USA</i> | Jun. 2026 |
| 2. Louisiana State University Health Sciences
<i>New Orleans, LA, USA</i> | Jun. 2026 |
| 3. Stony Brook University School of Medicine
<i>Stony Brook, NY, USA</i> | Jun. 2026 |
| 4. Cold Spring Harbor Laboratory (virtual seminar)
<i>Cold Spring Harbor Laboratory, NY, USA</i> | May 2026 |
| 5. Fox Chase Cancer Center
<i>Philadelphia, PA, USA</i> | May 2026 |
| 6. Purdue University Department of Biological Sciences
<i>West Lafayette, IN, USA</i> | Mar. 2026 |
| 7. University of California, San Francisco (Pharm. Sci.)
<i>San Francisco, CA, USA</i> | Mar. 2026 |
| 8. UT Austin – Oden Institute & Dell Med. School
<i>Austin, TX, USA</i> | Mar. 2026 |
| 9. University of Notre Dame Dept. of Chemistry
<i>South Bend, IN, USA</i> | Dec. 2025 |
| 10. University of Maryland School of Medicine
<i>Baltimore, MD, USA</i> | Dec. 2025 |
| 11. Lund University Faculty of Medicine
<i>Lund, Skåne, Sweden</i> | Oct. 2025 |
| 12. CSIR: Institute for Genomics and Integrative Biology
<i>New Delhi, India</i> | Aug. 2025 |
| 13. Laufer Center at Stony Brook University
<i>Stony Brook, NY, USA</i> | Apr. 2025 |
| 14. Biophysical Society 2025 Platform Talk (Co-Chair of session)
<i>Los Angeles, CA, USA</i> | Feb. 2025 |

15. NYU Simons Center for Computational Physical Chemistry <i>New York University, New York, NY, USA</i>	<i>Oct. 2024</i>
16. Gordon Research Seminar: “Phosphorylation and G-protein mediated signaling networks” <i>Waterville Valley, NH, USA (location of GRC/GRS)</i>	<i>June 2024</i>
17. Illinois Institute of Technology Dept. of Chemistry <i>Chicago, IL, USA</i>	<i>May 2024</i>
18. American Chemical Society COMP division invited seminar <i>New Orleans, LA, USA</i>	<i>Mar. 2024</i>
19. National Cancer Institute Junior Investigators Meeting <i>University of Pennsylvania, Philadelphia, PA, USA</i>	<i>Aug. 2023</i>
20. School of Chemistry, University of Edinburgh <i>U. of Edinburgh, Edinburgh, UK</i>	<i>July 2023</i>
21. CCPBioSim 2023 – Biomolecular Simulations for a Better World <i>U. of Leeds, Leeds, UK</i>	<i>July 2023</i>
22. Single Cell Analysis and Innovation Lab, MSKCC <i>New York, NY, USA</i>	<i>Feb. 2022</i>
23. RockEDU Teacher’s Summit <i>New York, NY, USA</i>	<i>Jan. 2023</i>
24. SciLifeLab Stockholm <i>Stockholm, Sweden</i>	<i>Oct. 2022</i>
25. Loyola High School, New York, NY <i>Loyola High School in Manhattan, NY</i>	<i>Apr. 2021</i>
26. Biophysical Society 2020 Annual Meeting <i>San Diego, CA, USA</i>	<i>Nov. 2020</i>
27. MilliporeSigma Fellowship lecture <i>St. Louis, MO, USA</i>	<i>Nov. 2020</i>
28. Protein Folding Consortium 2019 <i>St. Louis, MO, USA</i>	<i>Feb. 2020</i>
29. Gibbs Conference in Biothermodynamics <i>Carbondale, IL, USA</i>	<i>Nov. 2019</i>
30. Biochemistry and Molecular Biophysics Science Friday Seminar <i>St. Louis, MO, USA</i>	<i>July 2019</i>
31. Department of Chemistry Capstone seminar <i>St. Louis, MO, USA</i>	<i>October 2017</i>
32. Midstates Consortium of Math and Sciences <i>Chicago, IL, USA</i>	<i>October 2016</i>

Media Appearances

Youtube – Folding@home <i>What happens when you run a work unit on your PC using Folding@home?</i> Available on Youtube	<i>Dec. 2020</i>
Bloomberg Government <i>‘Creativity Is Simply Lost’ as Covid Cripples Academic Research. Article Link</i>	<i>Oct. 15, 2020</i>

St. Louis Post Dispatch

June 26, 2020

Gamers, Big Tech, even La Liga soccer link computers to fight COVID-19.[Article Link](#)**Association for Computing Machinery – SIGGRAPH**

May 28, 2020

Folding@home: Citizen Scientists Gain Insight on COVID-19. [Article link](#)**Peer-
Reviews
conducted
for scientific
journals**

ACS Chemical Biology
Nature Computational Sciences
Journal of Medicinal Chemistry
Biophysical Journal
Journal of Chemical Theory and Computation
Journal of Chemical Information and Modeling
eLife
Proteins
PLOS One
Protein Science
Living Journal of Computational Molecular Sciences

**Teaching
Experience****New York University**

Fall 2023

*Guest lecture: Computational Physical Chemistry***Illinois Inst. of Technology**

Spring 2024

*Guest lecture: Computational Biochemistry***Weill Cornell Medical School**

Spring 2025

*Guest lecture: Deep learning in biology***Teaching Assistant: General Biochemistry**

Aug 2015 – Dec 2015

*Dept. of Biology, Washington University in St. Louis***Teaching Assistant: Modern Medicinal Chemistry**Jan. 2014 – June
2014*Dept. of Chemistry, Washington University in St. Louis***Lab instructor: Introductory Organic Chemistry**June 2014 – Aug.
2013*Dept. of Chemistry, Washington University in St. Louis***Teaching assistant: Introduction to Computer Science**

Aug. 2010 – Dec 2011

Dept. of Computer Science, Washington University in St. Louis